

A review of graph neural networks and pretrained language models for knowledge graph reasoning

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Introduction & Background

- Knowledge graphs represent entities and relationships as nodes and edges
- Facts are encoded as triples (head, relation, tail)
- KGR infers missing facts and answers queries
- Tasks include knowledge graph completion & question answering
- Approaches: symbolic, embedding-based & hybrid

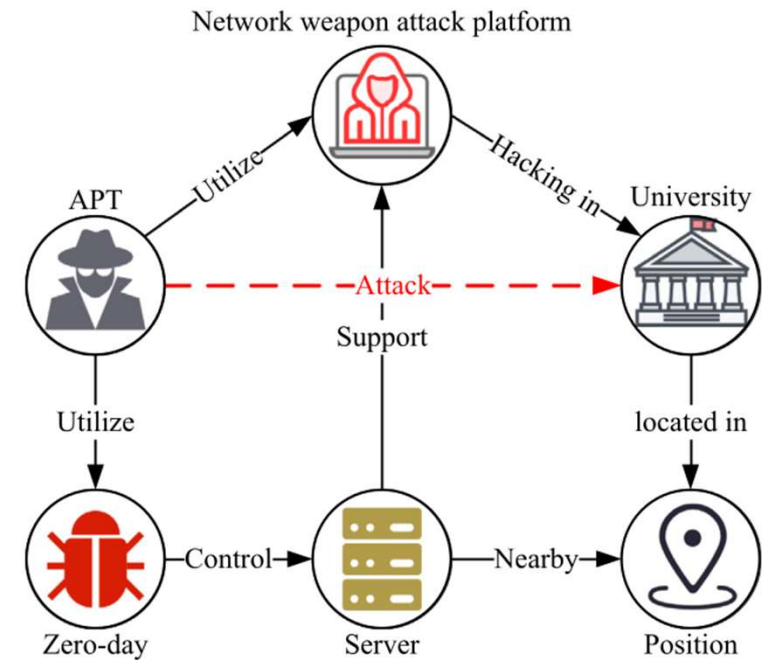


Fig. 2. Illustration of a cybersecurity KG and examples of two reasoning tasks. (1) KGC: Query the missing relation between head entity "APT" and tail entity "University". The reasoning result is "Attack". (2) KGQA: Who attacked the "University" using the "Network weapon attack platform"? The reasoning result is "APT".

General KGR Framework

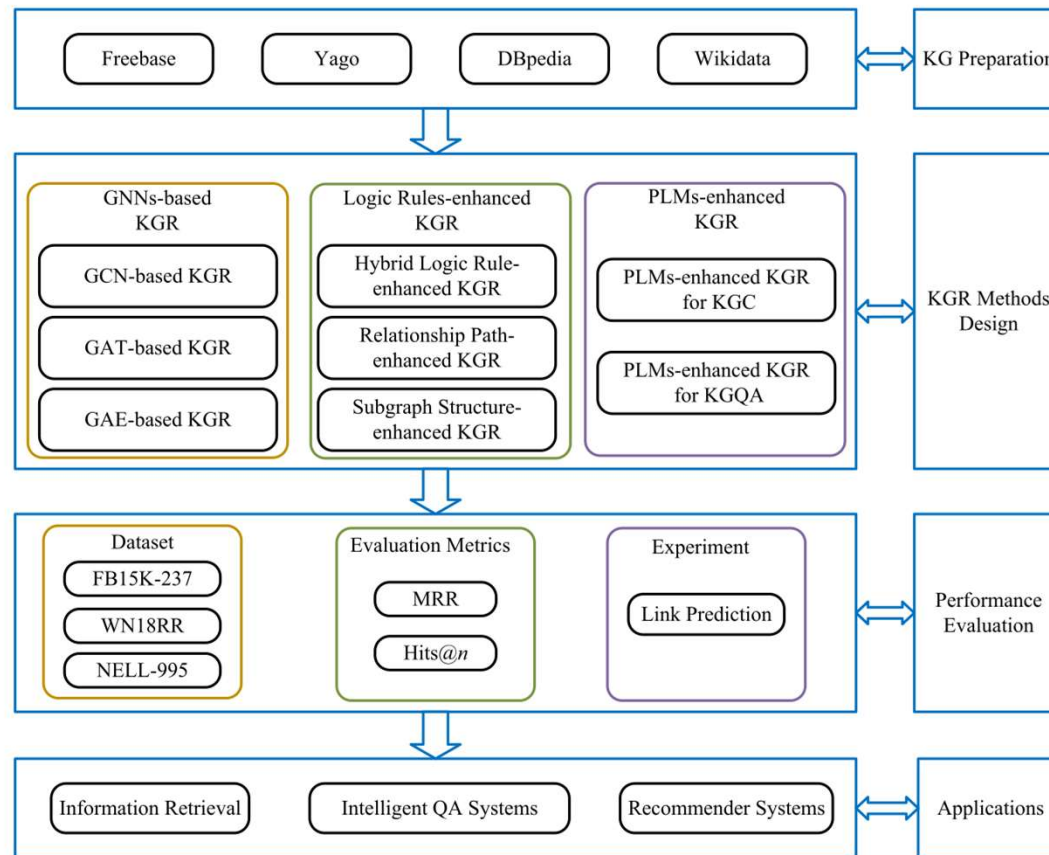


Fig. 1. The general framework of KGR consists of four main steps: (1) KG preparation, (2) KGR methods design, (3) Performance evaluation, and (4) Applications.

Method Categories

GNN-Based

- Operate on graph structure using convolution, attention or autoencoder
- Capture local and global patterns
- Variants: GCN, GAT, GAE

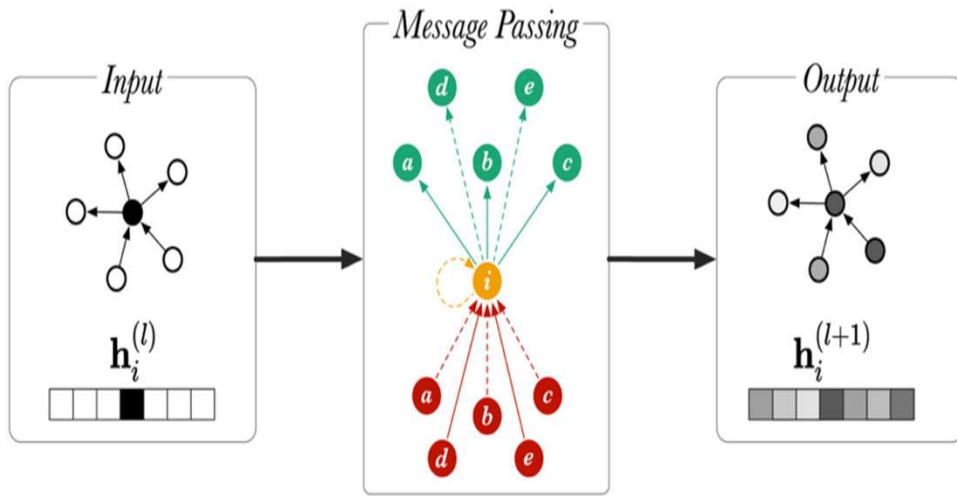
Logic Rules

- Use formal logic
- Enhance GNNs via rules & relation paths
- Improve interpretability & generalization

PLM-Enhanced

- Leverage PLMs for rich semantic context
- Bridge text & graph modalities
- Categories: KG-enhanced LLMs & LLM-augmented KGs

A normal GNN architecture



data edges



incoming edges



outgoing edges

added edges



inverse edges
of incoming edges



inverse edges of
outgoing edges



self-looping edges

Message passing layer.

Input: $A, \sigma, [X]$

Result: H

if *featureless* **then**

$x \leftarrow \text{einsum}(ni, io \rightarrow no, Ah, W)$

$H \leftarrow \sigma(x)$

else

if *horizontally stacked* **then**

$x \leftarrow \text{einsum}(ni, rio \rightarrow rno, X, W)$

reshape x into a matrix with the dimensions $NR^+ \times N_{out}$

$x \leftarrow \text{spmm}(A_h, x)$

else

$x \leftarrow \text{spmm}(A_v, X)$

reshape x into a tensor with the dimensions $R^+ \times N \times N_{in}$

$x \leftarrow \text{einsum}(rio, rni \rightarrow no, W, x)$

end

$H \leftarrow \sigma(x)$

end

1. Graph Convolutional Networks (GCN):

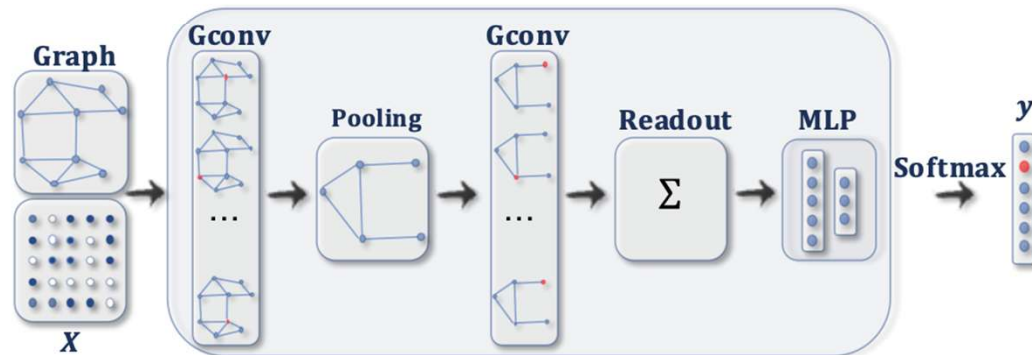
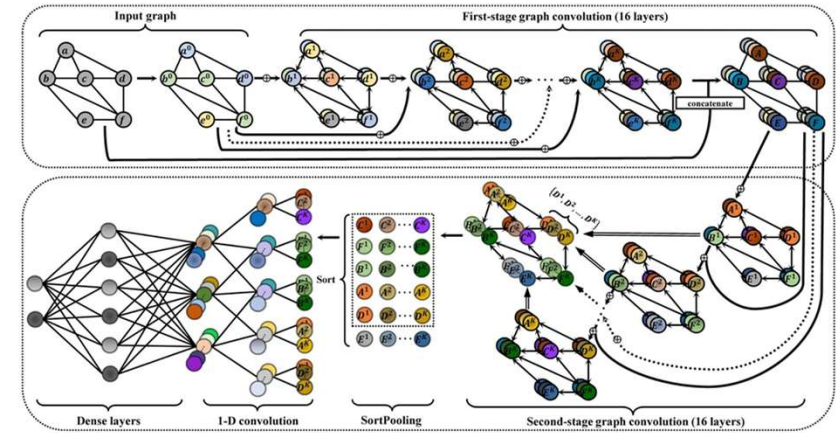
GCNs are one of the most popular GNN architectures. They extend the concept of convolutions to graph data by aggregating information from neighboring nodes. The core idea is to compute a node's new representation by aggregating the features of its neighbors, weighted by the graph's structure (adjacency matrix).

- **Formula:**

$$h_i^{(k+1)} = \sigma \left(\sum_{j \in N(i)} \frac{1}{C_{ij}} W^{(k)} h_j^{(k)} \right)$$

Where:

- $h_i^{(k)}$ is the representation of node i at the k -th layer.
- $N(i)$ represents the neighbors of node i .
- C_{ij} is a normalization constant (degree of the node or more complex normalizations).
- $W^{(k)}$ is the learned weight matrix.



2. Graph Attention Networks (GAT):

GATs enhance GCNs by introducing attention mechanisms to assign different importance to neighboring nodes, instead of treating them equally (like GCNs do). This is done by using attention coefficients that dynamically learn how much influence each neighbor has on the node's new representation.

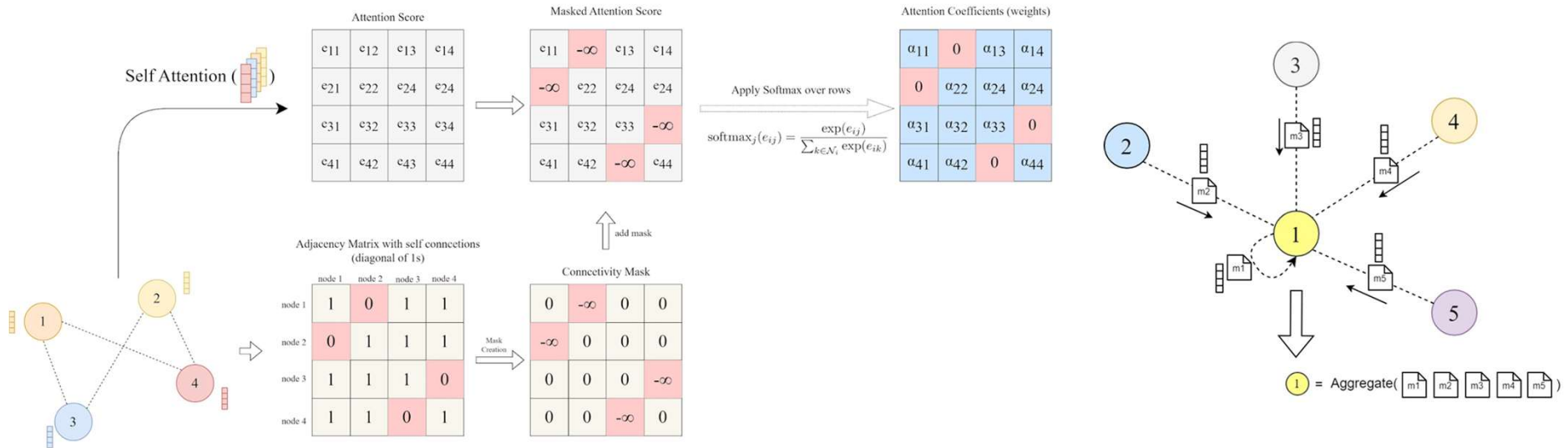
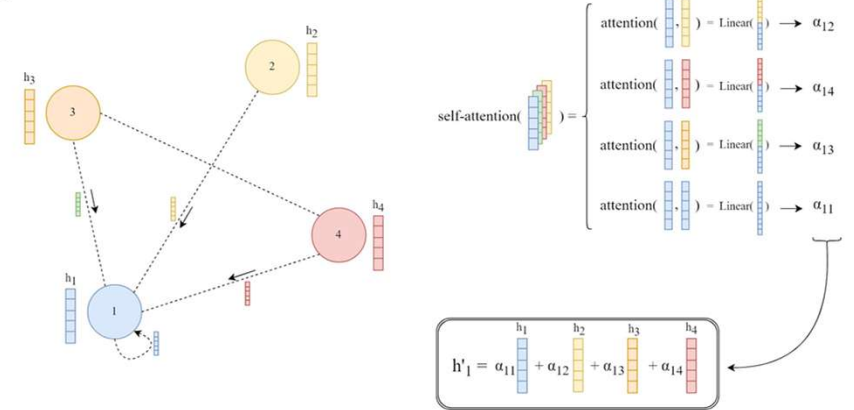
- **Formula:**

$$h_i^{(k+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} W^{(k)} h_j^{(k)} \right)$$

Where:

- α_{ij} is the attention coefficient between nodes i and j , typically computed using a self-attention mechanism (like scaled dot-product attention).

Graph Attention Networks

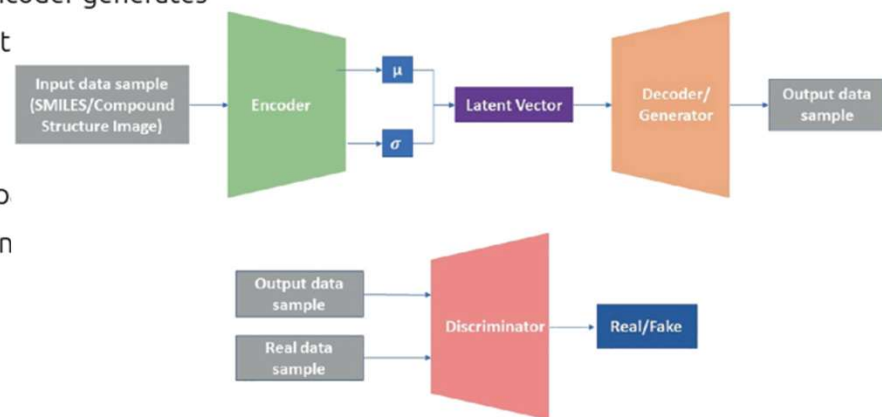


3. Graph Autoencoders (GAE):

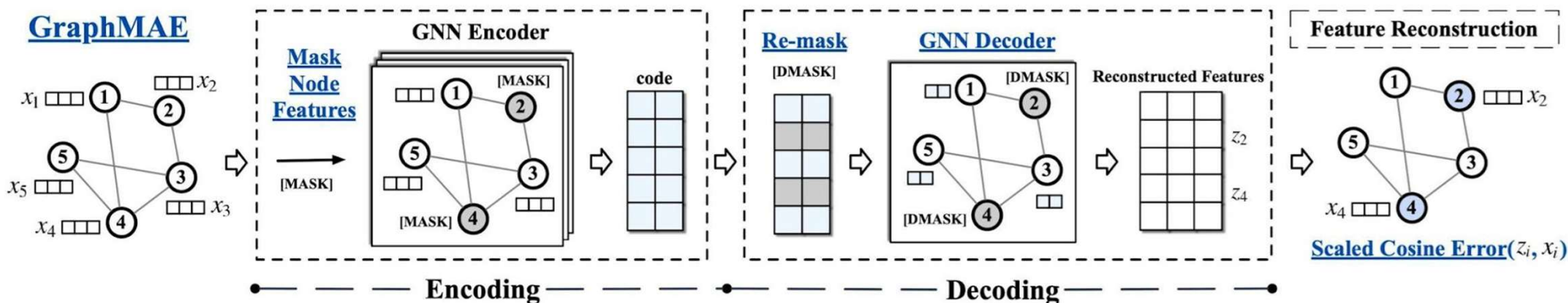
GAEs are unsupervised learning models that aim to learn graph embeddings, focusing on reconstructing the graph structure. The model typically uses an encoder-decoder framework where the encoder generates low-dimensional representations of the graph (node embeddings), and the decoder reconstructs (usually predicting missing edges).

- **Structure:**

- **Encoder:** Uses a GCN or similar model to embed nodes into a lower-dimensional space
- **Decoder:** Reconstructs the graph from the embeddings (e.g., predicting missing link nodes).



GraphMAE



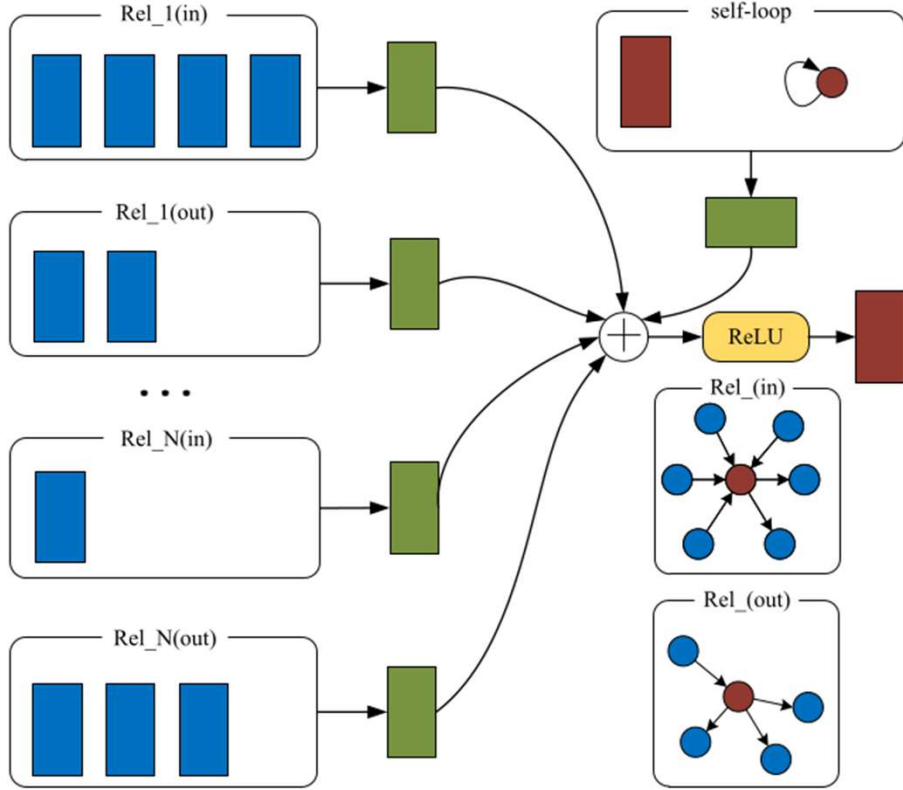


Fig. 3. Illustration of the model structure of R-GCN [48]. The red rectangle represents the information of the red node that RGCN is currently updating. The blue nodes are neighbors of the red nodes, the blue rectangle represents their feature vectors. The normalization results (green rectangle) arise from the matrix multiplication of the feature vectors of the blue nodes with a specific relational transformation matrix. The red node's feature vectors are updated through the summation of the normalization results and the application of the ReLU activation function.

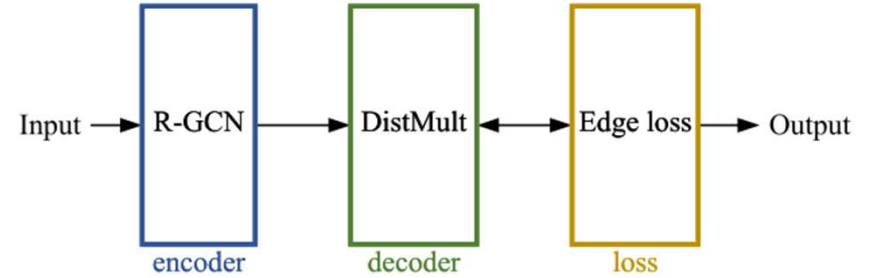
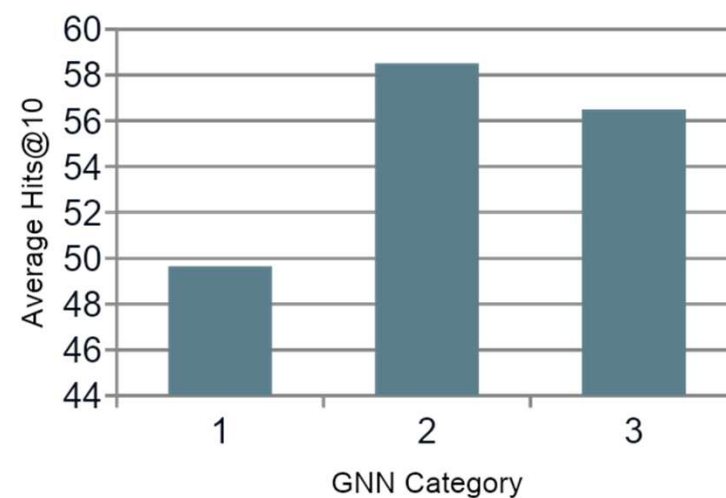


Fig. 4. Illustration of the model structure of encoder-decoder [48]. R-GCN obtains the embedding representations of the nodes in the input KG. DistMult receives the embedding representations of the nodes from the output of R-GCN and predicts the edges in the KG based on these embedding representations. Edge Loss optimizes the model parameters based on the losses to improve the accuracy of the prediction and outputs the final evaluation metrics along with the corresponding prediction results.

GNN-Based Reasoning

Category	Key Idea	Strengths	Limitations
GCN	Convolution aggregates neighbourhood	Captures local structure; efficient	Transductive; over-smoothing
GAT	Attention weights neighbours	Focuses on important nodes	High complexity; memory
GAE	Autoencoders reconstruct graph	Learns global patterns	Complex training; unsupervised



GAT-based models achieve the best mean Hits@10 on FB15k-237

Table 1

The classification and features analysis of GNNs-based KGR methods.

Type	Model	Year	Features analysis
GCN	R-GCN [48]	2018	Extending GCN to directed multi-relational graphs
	SACN [49]	2019	End-to-end web framework scales to large-scale KGs
	CompGCN [50]	2020	Combining node characterization updates and relationship integration
	KE-GCN [51]	2021	Combining various Knowledge Graph Embedding (KGE) methods into GCN
	REGCN [52]	2021	Cyclical modeling of KG sequences to learn evolutionary representations of entities and relationships
	CEN [53]	2022	Encoding KG sequences with R-GCN and decoding the evolving representation with length-aware convolutional networks
	RPC [54]	2023	Leverage relational correlations and periodic events in TKG to improve model representation and inference performance
GAT	GAT [55]	2018	Computationally efficient, interpretable, and powerful representations
	KBGAT [56]	2019	Strong ability to handle multiple and long-distance relationships
	DPMPN [57]	2020	Constructing graph-structured explanations to enhance reasoning and interpretability
	RGHAT [58]	2020	Increasing the interpretability of models using hierarchical attention mechanisms
	RAGAT [59]	2021	Utilise a larger sliding window and consider more relational triples
	DisenKGAT [60]	2021	Enhanced model representation using decoupled graph attention networks with strong robustness
	TAGAT [61]	2021	High reasoning performance and interpretability using hierarchical attention mechanism to perceive type and neighbor information
GAE	HGAT [62]	2023	Design time-aware and relationship-aware attention mechanisms to learn more valuable information from relevant historical facts
	GAE/VGAE [63]	2016	Simultaneous learning of graph structure information and node content features is possible
	DRNE [64]	2018	Learning network embeddings with rule-equivalent nodes better captures structural information about the graph
	NetGAN [65]	2018	Important topological information is preserved and link prediction is better
	M-GNN [66]	2019	Dynamically learns different subgraph structures and adaptively adapts
	M ² GNN [67]	2021	Adaptive learning of the importance between layers enhances embedding
	SeeGera [68]	2023	Learn node representation by reconstructing input graph data
	ProtoMGAE [69]	2024	Integrated contrast learning and automatic encoder of masking graph, with strong representation ability of learning graph nodes
S2GAE [70]		2023	The representation of nodes is learned by reconstructing the structure of the graph, which is highly extensible

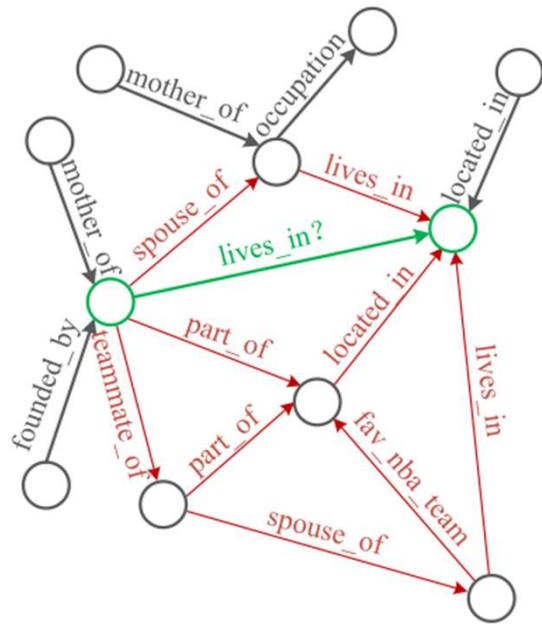
Logic Rule-Enhanced Reasoning

- Symbolic rules encode domain knowledge (Horn clauses, Prolog, ASP)
- Hybrid models integrate rules with GNNs
- Enhances interpretability & leverages global patterns
- Challenges: rule learning is laborious & scalability

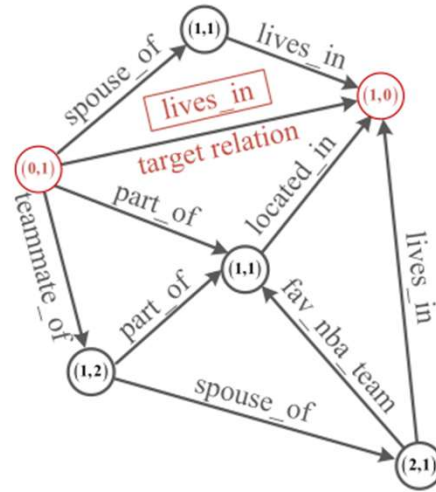
Rule Base

GNN Model

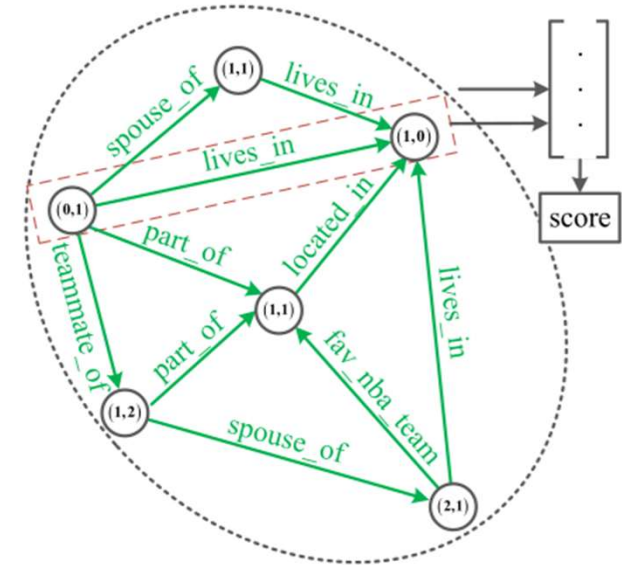
Reasoning Output



(a) Subgraph extraction



(b) Node labeling



(c) GNN scoring

Fig. 5. Illustration of model structure of GraIL [100]. (a) Subgraph extraction: extract the closed subgraph of the red part surrounding the two green target nodes. (b) Node labeling: labeling the nodes in the extracted closed subgraph with features and providing the relevant structure information of the graph to GNN, where the number represents the shortest path length from the current node to the two target nodes. (c) GNN scoring: use GNN for message passing, and score the labeled subgraph.

Table 2

The classification and features analysis of logic rules-enhanced KGR methods.

Type	Model	Year	Features analysis
Rule	VN [90]	2020	Very robust to sparsity problems and able to embed invisible entities more accurately
	pGAT [91]	2020	Enhancing KGE performance using domain information
	ExpressGNN [92]	2020	Effectively handle large-scale probabilistic logic rules and KGs to support complex reasoning tasks
	DA-GCN [93]	2022	Using logic rules to enhance generalized types of GNNs and reduce data dependence
	GNN-QE [94]	2022	Strong generalization performance for more interpretable multi-hop reasoning
Path	NBFNet [95]	2021	Solving path formulas through the Bellman-Ford algorithm to effectively improve path formulation
	INDIGO [96]	2021	By leveraging the semantic information of entities, enhance the accuracy of KGR
	CBGNN [97]	2022	Efficiently learning loops that represent good rules through a given loop base can help ternary prediction
	RED-GNN [98]	2022	Using relational directed graphs to capture structural information of KGs and enhance reasoning interpretability
	AdaProp [99]	2023	Adaptive tuning based on entity representations, efficient and able to consider unknown target entities
Subgraph	GraIL [100]	2020	Inductive reasoning based on local subgraph structure enables the model to generalize to unseen entities outside of the training data
	TACT [101]	2021	Leveraging topological perception and correlation between relationships for inductive KGR
	CoMPiLE [102]	2021	Using directed subgraphs to infer the relationship between two invisible entities
	Meta-iKG [103]	2023	Using meta-learning and local subgraph-based few-shot KGR
	CSR [104]	2022	CSR predicts directly on the target small-sample task without pre-training on the training task set
	ConGLR [105]	2022	Reasoning using context graph-based relationship embedding and relationship path representation
	SGI [106]	2022	Mutual information between a goal relation and its closed subgraph can be maximized
	SNRI [107]	2022	Enhancing reasoning by using adjacency features of nodes and adjacency paths of sparse subgraphs
	LogCo [108]	2022	Integrating neural networks and discrete logic and contrastive representations into inductive relational prediction for interpretability of reasoning
	DEKG-ILP [109]	2023	Extension of an existing inductive link prediction formula for invisible entities in disconnected emergent scenarios
	InGRAM [110]	2023	Generate embedding vectors for new relations and entities that appear only at reasoning time

PLM-Enhanced Reasoning

- Pre-trained language models provide rich contextual semantics
- KG-enhanced LLMs improve factual grounding & reduce hallucination
- LLM-augmented KGs assist construction, completion & querying
- Bridging textual & structural modalities enhances reasoning



Table 3

The features analysis of PLMs-enhanced KGR methods for the KGC task.

Model	PLM	Year	Features analysis
KG-BERT [123]	BERT	2019	First proposal of using PLM for KGC
MTL-KGC [124]	BERT	2020	PLM based multitask learning method for KGC
StAR [125]	BERT&RoBERTa	2021	Combining graph structure coding with text coding to synthesize structural and textual information
KEPLER [126]	RoBERTa	2021	Joint training of knowledge embedding and MLM objectives on the same PLM
InductiveE [127]	BERT	2021	Proposing an inductive learning model for commonsense KGC and constructing a representation of invisible entities
BLP [128]	BERT	2021	Using PLM with fine-tuning through link prediction
SimKGC [129]	BERT	2022	Introducing contrastive learning to improve text-based embedding methods and graph-based rearrangement strategies to capture topological information between entities in KGs
PKGC [130]	RoBERTa	2022	Each triad and its supporting information is converted into natural cue sentences, which are then further entered into the PLM for categorization
BERTRL [131]	BERT	2022	BERTRL uses reasoning paths and triples as input to PLMs, which can process unseen entities and gain relational reasoning capability
LMKE [132]	BERT	2022	Using LM to achieve knowledge embeddings and implementing negative sampling for via contrastive learning

Table 4

The features analysis of PLMs-enhanced KGR methods for the KGQA task.

Model	PLM	Year	Features analysis
KagNet [133]	BERT	2019	Using the KG as an external knowledge source, GCN and path-based hierarchical attention mechanisms are introduced to compute relevance scores
MHGRN [134]	BERT&RoBERTa	2020	Multi-hop reasoning question and answer task by extending R-GCN direct encoding paths with a structured relational attention mechanism
QA-GNN [135]	RoBERTa	2021	Add question and answer contexts as nodes to the subgraph to increase the interaction of problem contexts with the KG
DRAGON [136]	RoBERTa	2022	Propose KG subgraph correlation dynamic neighborhood matrix to update node representation based on node correlation
GrapeQA [137]	RoBERTa	2023	Proposed graph enhancement and pruning strategies can improve the information flow between question and answer and KG by pruning less relevant information
JointLK [138]	RoBERTa	2022	Fine-grained modal fusion and multi-layer joint reasoning between language models and KGs
UniKGQA [139]	RoBERTa	2023	Proposing pre-training and fine-tuning strategies to solve multi-hop KGQA Tasks
DHLK [140]	RoBERTa	2023	Constructing heterogeneous knowledge graph (HKG) that deeply integrates both QA context and HKG.
GREASELM [141]	RoBERTa	2022	Fusion of coded representations from PLMs and GNNs on multilayer modal interaction operations
FiTs [142]	RoBERTa	2023	Aligning PLM and KG representations to bridge the modal gap between them and improve joint reasoning based on the aligned representations
MRGNN [143]	BERT&RoBERTa	2024	Proposed meta-path reasoning GNN that aligns meta-paths and question-answer contexts and employs multilevel attention graphs to enhance the information interaction between language models and KGs

Evaluation Metrics & Datasets

Mean Reciprocal Rank (MRR)

$$MRR = \frac{1}{2|T_{test}|} \sum_{i \in T_{test}} \left(\frac{1}{rank_i^h} + \frac{1}{rank_i^t} \right)$$

Hits@K

$$Hits@n = \frac{1}{2|T_{test}|} \sum_{i \in T_{test}} I[rank_i^h \leq n] + I[rank_i^t \leq n]$$

Proportion of predictions where the correct answer appears within top K

Dataset	Entities	Rel.	Train	Valid	Test
FB15k-237	14,541	237	272,115	17,535	20,466
WN18RR	40,943	11	86,835	3,034	3,134

Results

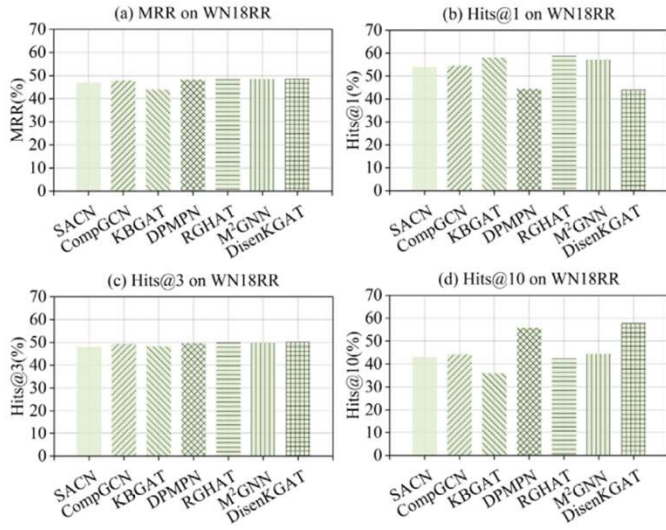


Fig. 7. The transductive model's link prediction results on the WN18RR dataset.

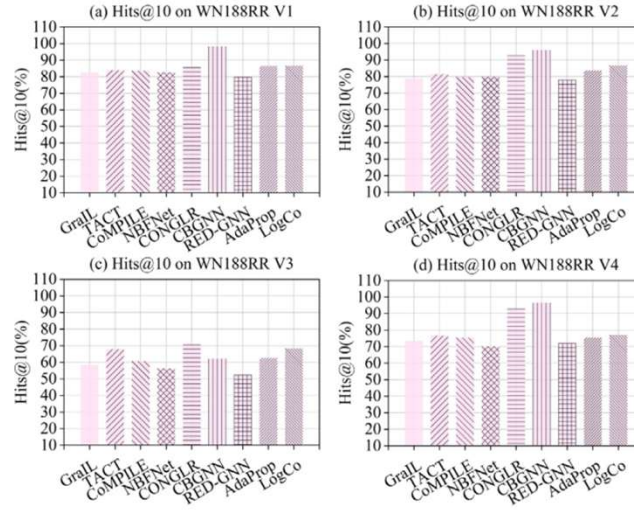


Fig. 8. The inductive model's link prediction results on the WN18RR V1-V4 dataset.

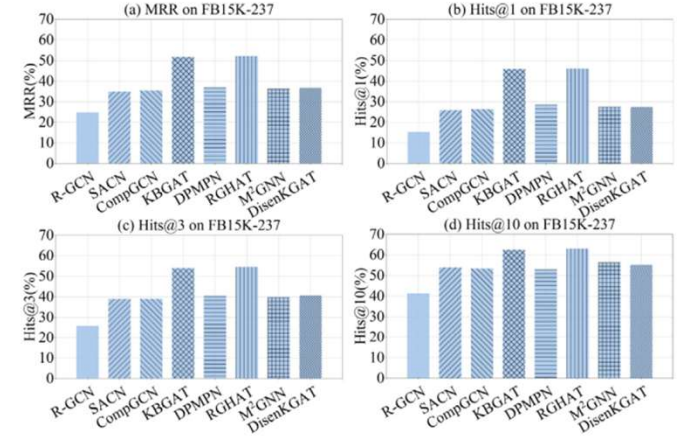


Fig. 6. The transductive model's link prediction results on the FB15K-237 dataset.

Table 7

The transductive model’s link prediction results (%)

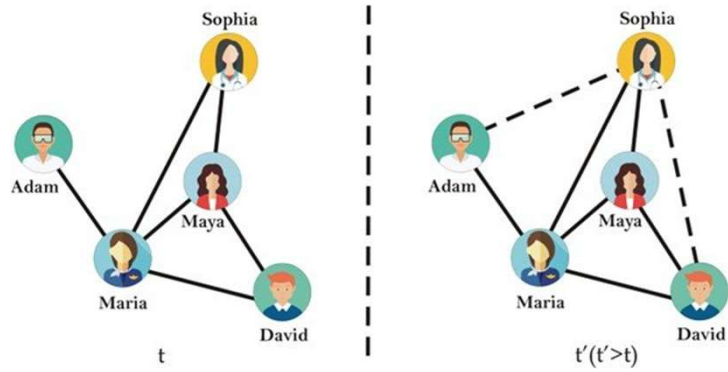
Model	Fb15k-237				WN18RR			
	MRR	Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10
R-GCN [48]	24.8	15.3	25.8	41.4	–	–	–	–
SACN [49]	35	26	39	54	47	43	48	54
CompGCN [50]	35.5	26.4	39	53.5	47.9	44.3	49.4	54.6
KBGAT [56]	51.8	46	54	62.6	44	36.1	48.3	58.1
DPMPN [57]	36.9	28.6	40.3	53	48.2	44.4	49.7	55.8
RGHAT [58]	52.2	46.2	54.6	63.1	48.3	42.5	49.9	58.8
M ² GNN [67]	36.2	27.5	39.8	56.5	48.5	44.4	49.8	57.2
DisenKGAT [60]	36.8	27.5	40.7	55.3	48.6	44.1	50.2	57.8

Table 8

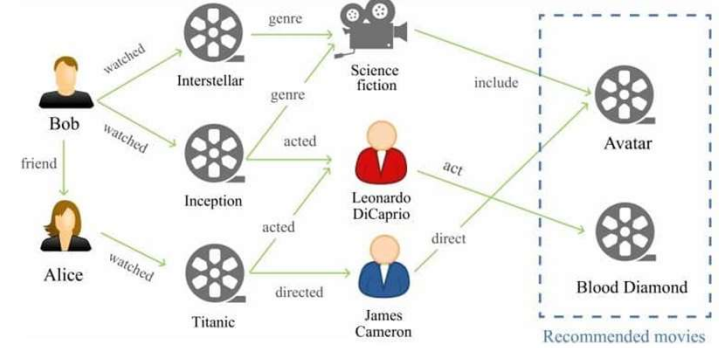
The inductive model’s link prediction results on Hits@10 (%).

Model	WN18RR				Fb15k-237				NELL-995			
	V1	V2	V3	V4	V1	V2	V3	V4	V1	V2	V3	V4
GraIL [100]	82.5	78.7	58.4	73.4	64.2	81.8	82.8	89.3	59.5	93.3	91.4	73.2
TACT [101]	84	81.6	68	76.6	65.8	83.6	85.2	88.7	79.8	88.9	94	73.8
CoMPiLE [102]	83.6	79.8	60.7	75.5	67.6	83	84.7	87.4	58.4	93.9	92.8	75.2
NBFNet [95]	82.7	79.9	56.3	70.2	51.7	63.9	58.8	55.9	79.5	63.5	60.6	59.1
CONGLR [105]	85.6	92.9	70.7	92.9	68.3	86	88.6	89.3	81	94.9	94.4	81.6
CBGNN [97]	98.4	96.1	62.3	96.5	97.6	96	94.9	94.7	84	95	95.3	92.3
RED-GNN [98]	79.9	78	52.4	72.1	48.3	62.9	60.3	62.1	86.6	60.1	59.4	55.6
AdaProp [99]	86.6	83.6	62.6	75.5	55.1	65.9	63.7	63.8	88.6	65.2	61.8	60.7
LogCo [108]	86.7	86.7	68.4	77.2	70.2	84.2	86.5	89.2	59.8	93.5	94.2	80.8

Use Case



Social Network



Recommender System



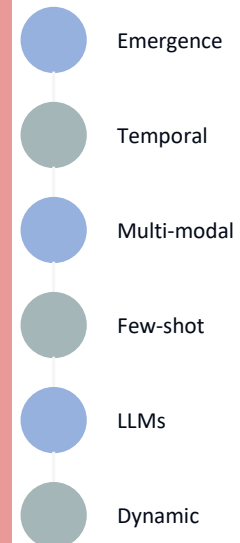
Cyber Security



Traffic System

Challenges & Future Directions

- Emergent entities & unseen relations
- Temporal & evolutionary patterns
- Multi-modal fusion & reasoning
- Few-shot & low-resource domains
- LLM integration & hybrid frameworks
- Dynamic KG maintenance & interpretability



The image features a solid yellow background. Overlaid on this background are three large, identical, light gray circles. These circles are arranged horizontally and overlap each other. The central circle is positioned slightly higher than the two flanking circles, creating a sense of depth. The text "Thank You" is centered within the middle circle.

Thank You